8. Moment of inertia:

Disc : $\frac{1}{2}mR^2$

Ring : mR^2

Hollow Sphere : $\frac{2}{3}MR^2$

Solid Sphere : $\frac{2}{5}MR^2$

Hollow Cone : $\frac{1}{2}MR^2$

Solid Cone : $\frac{3}{10}MR^2$

Thin Rod :

About Side $I = \frac{1}{12}ML^2$

AOR $I = \frac{1}{3}ML^2$

$I = mr^2$

RADIUS OF GYRATION:

$I = mK^2$

$K = \sqrt{\frac{I}{m}}$

i. Parallel Axis theorm. : $I = I_{COM} + Md^2$

ii. Perpendicular Axis theorm : $I_z = I_x + I_y$

$|\bar{L}| = \bar{P} \times r_{\text{Perpendicular}}$

$\bar{L} = \bar{r} \times m\bar{v}$. (∵ Angular Momentum)

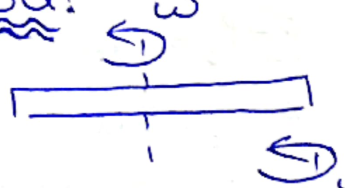
Torque ⇒ $\bar{\tau} = \bar{r} \times \bar{F}$

$\tau = r \cdot F\sin\theta$.

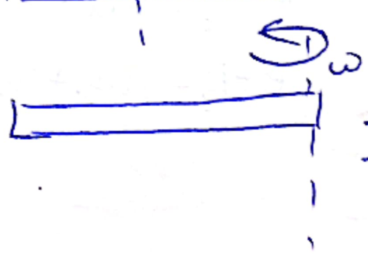
Where

$\theta \longrightarrow$ Angle b/w force, Distance Vector ...

Rod:



$I = \frac{1}{12}ML^2$



$I = \frac{1}{3}ML^2$